

Lecture 7: Geometry & 3D

Today:

00:00) Perspective vision

00:25) Homogeneous coordinates

00:45) ---- Break ----

01:00) Vanishing points

01:15) Transformations

01:45) ---- Break ----

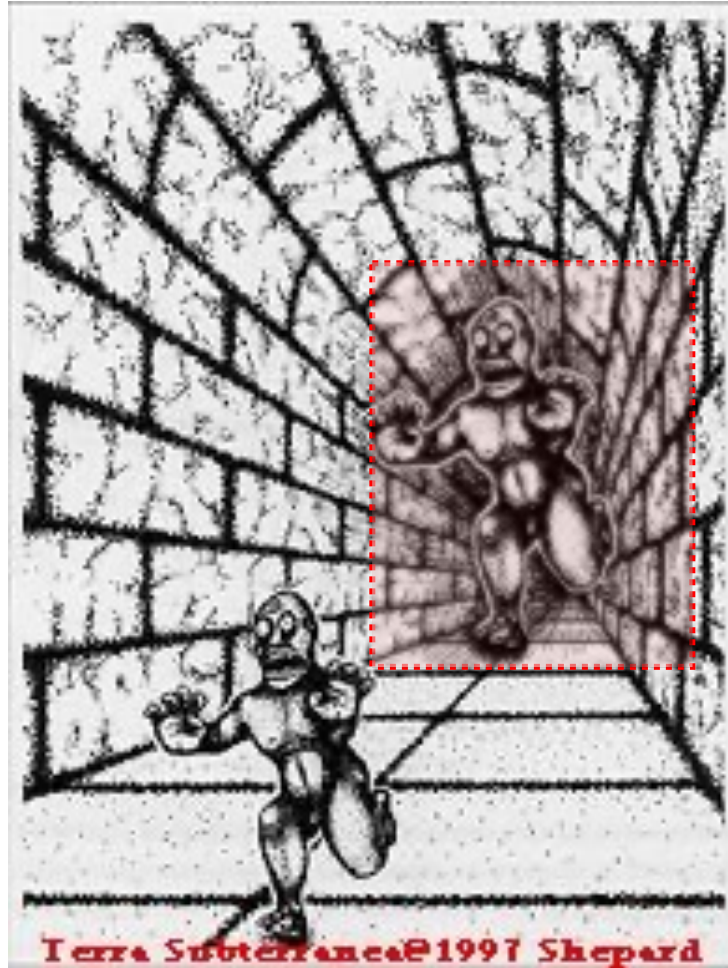
02:00) The camera matrix

02:20) Epipolar geometry

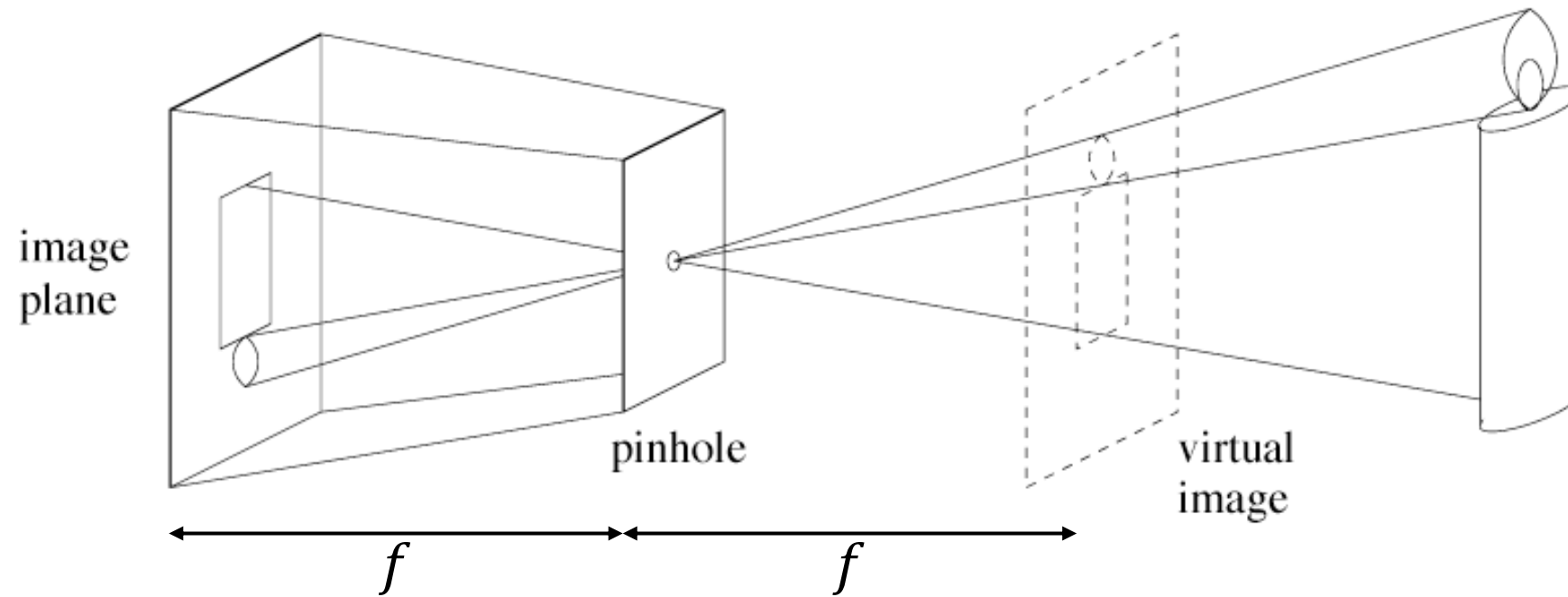
02:45) ---- Break ----

03:00) Tutorial: Deep 3D

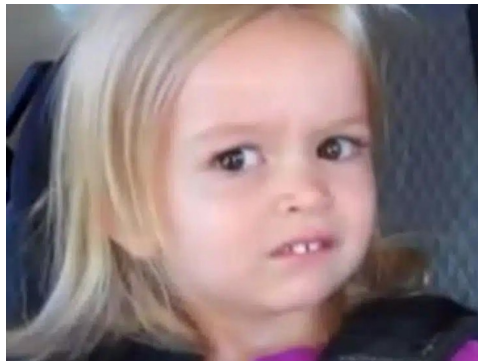
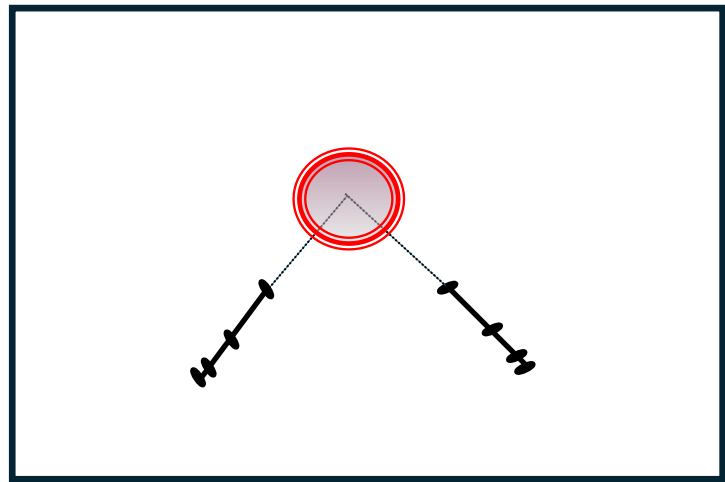
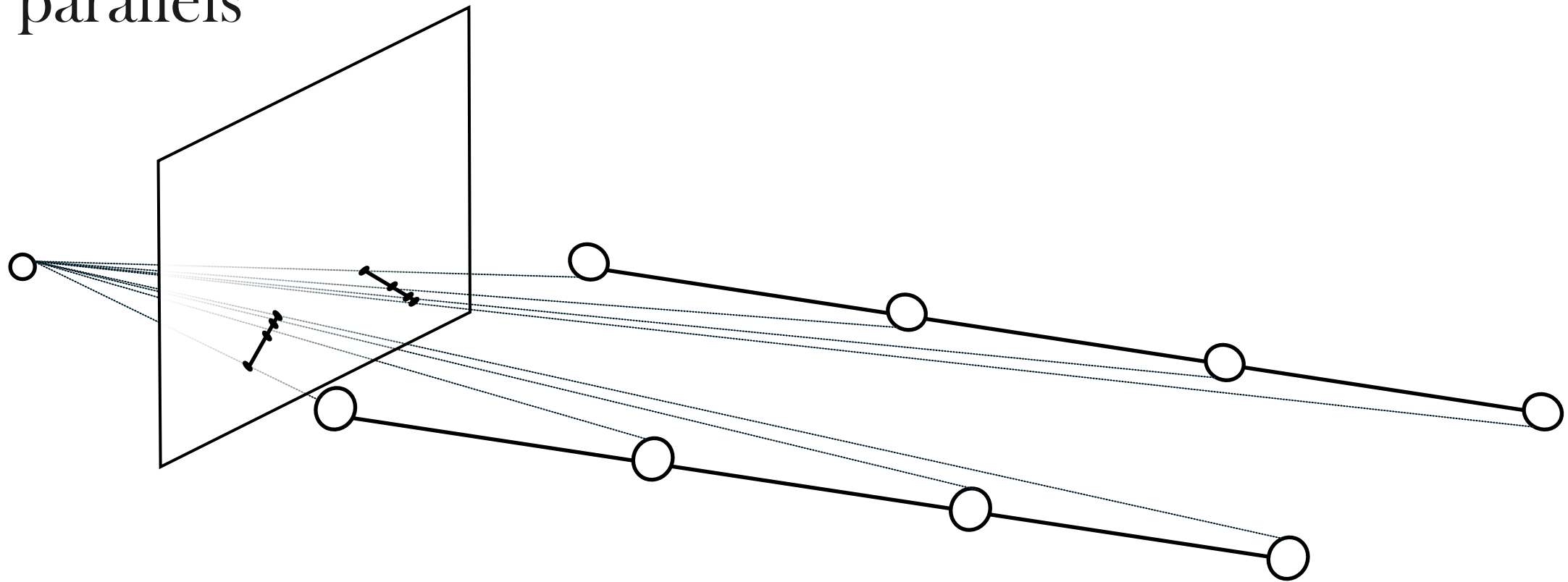
3D world projected to 2D image: Perspectivity



Pinhole camera model



Test-case: parallels



3rd graders already know!

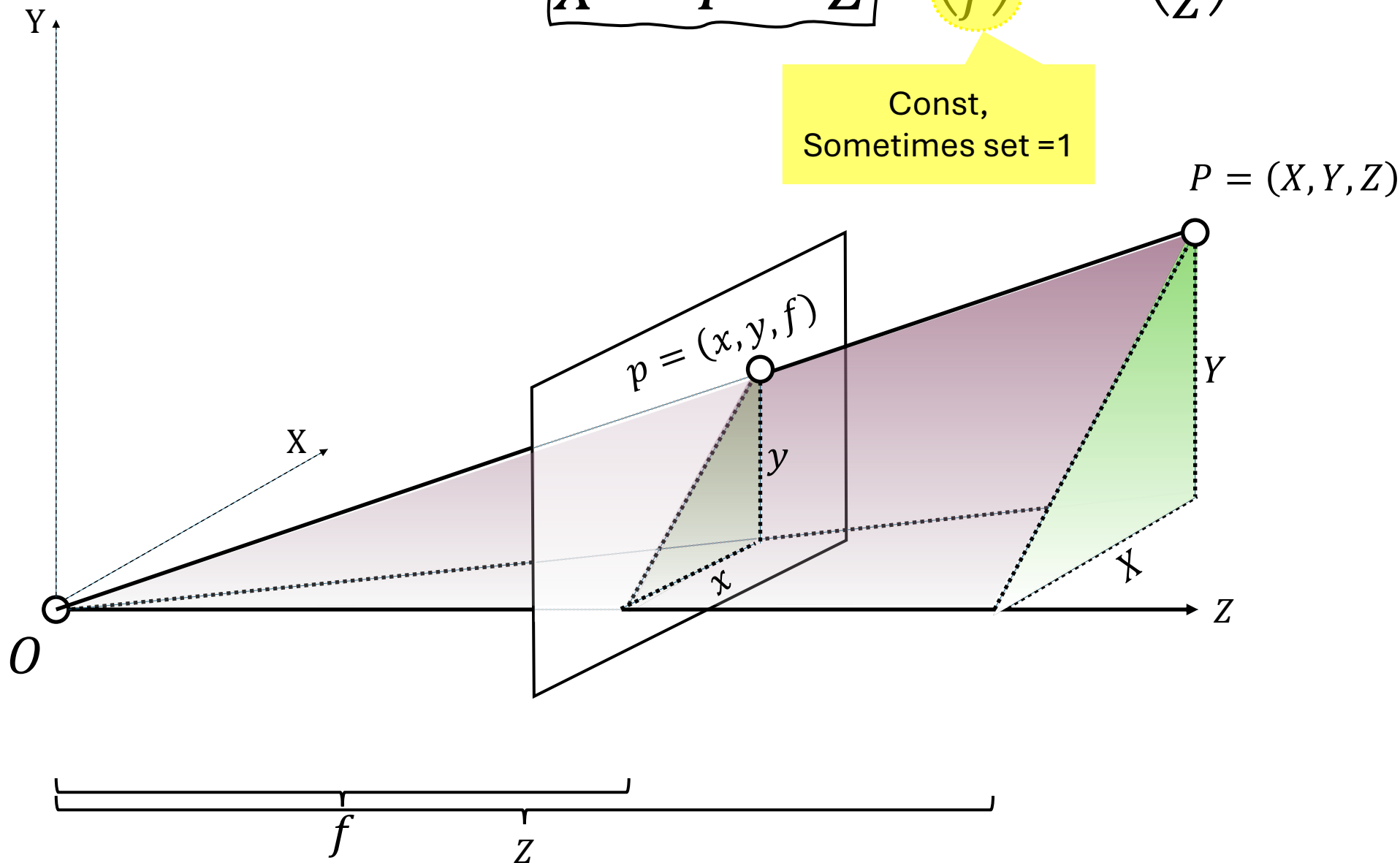


Credit: Plum class, "Pirhey Hamada" school, Rehovot.

Homogenous coordinates

$$\boxed{\frac{x}{X} = \frac{y}{Y} = \frac{f}{Z}} \quad \begin{pmatrix} x \\ y \\ f \end{pmatrix} = \frac{f}{Z} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \quad \text{Non-linear!}$$

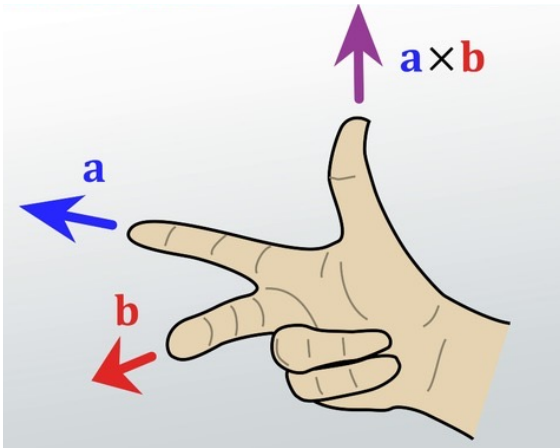
Const,
Sometimes set =1



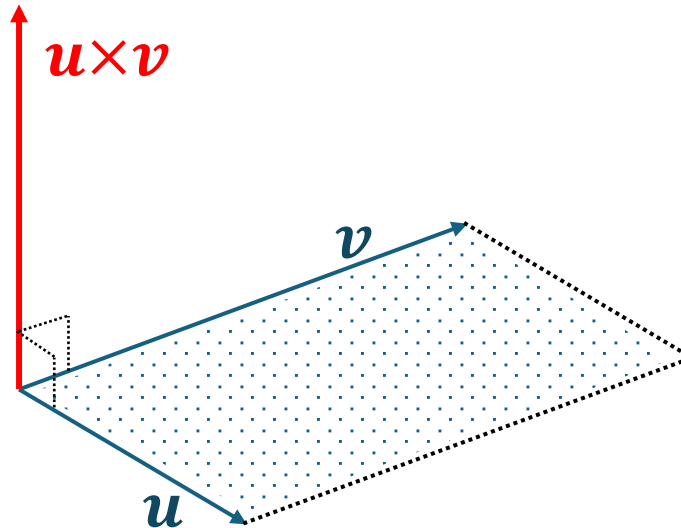
Recap: cross-product

$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}}_{[u]_{\times}} v = \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Orientation:
orthogonal to both



Magnitude: $\|u\| \cdot \|v\| \cdot \sin(\theta)$



More properties:

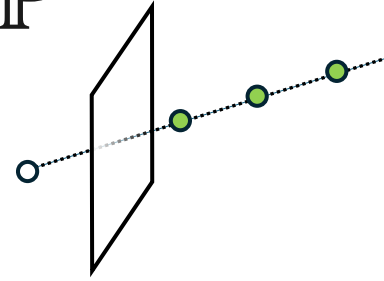
$$u \times v = -v \times u$$

$$a^T (b \times c) = 0 \Leftrightarrow \text{co-planar}$$

Homogenous coordinates properties $\mathbb{P}^2$

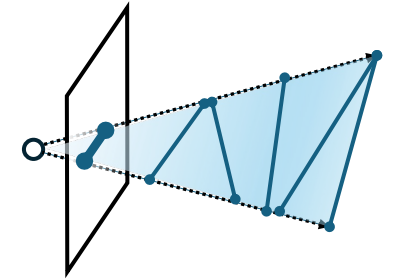
- $P \rightarrow p \iff \alpha P \rightarrow p$ ($\alpha \neq 0$).

Intuition:



$$\begin{pmatrix} x \\ y \\ 1 \\ p \end{pmatrix} = \frac{1}{Z} \begin{pmatrix} X \\ Y \\ Z \\ P \end{pmatrix}$$

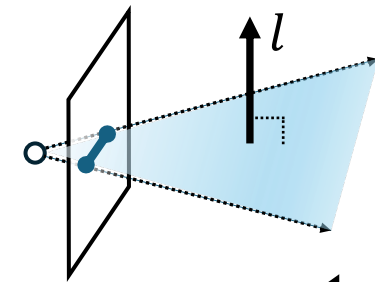
- A line in $\mathbb{P}^2$, $y = ax + b$, maps infinitely many lines in 3D: $\begin{pmatrix} xZ \\ (ax + b)Z \\ Z \end{pmatrix}$



- Lines map to lines. $aX + bY + cZ = 0 \implies a\frac{X}{Z} + b\frac{Y}{Z} + c = 0 \implies ax + by + c = 0$

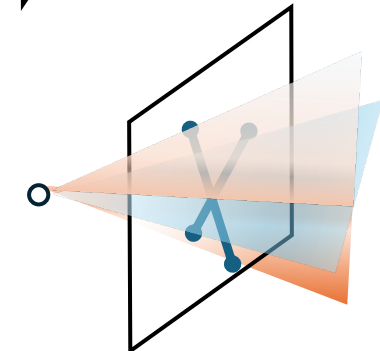
- Vectorized: $l^T P = 0 \implies l^T p = 0$.

l is a normal to the plane of all rays from origin to the line

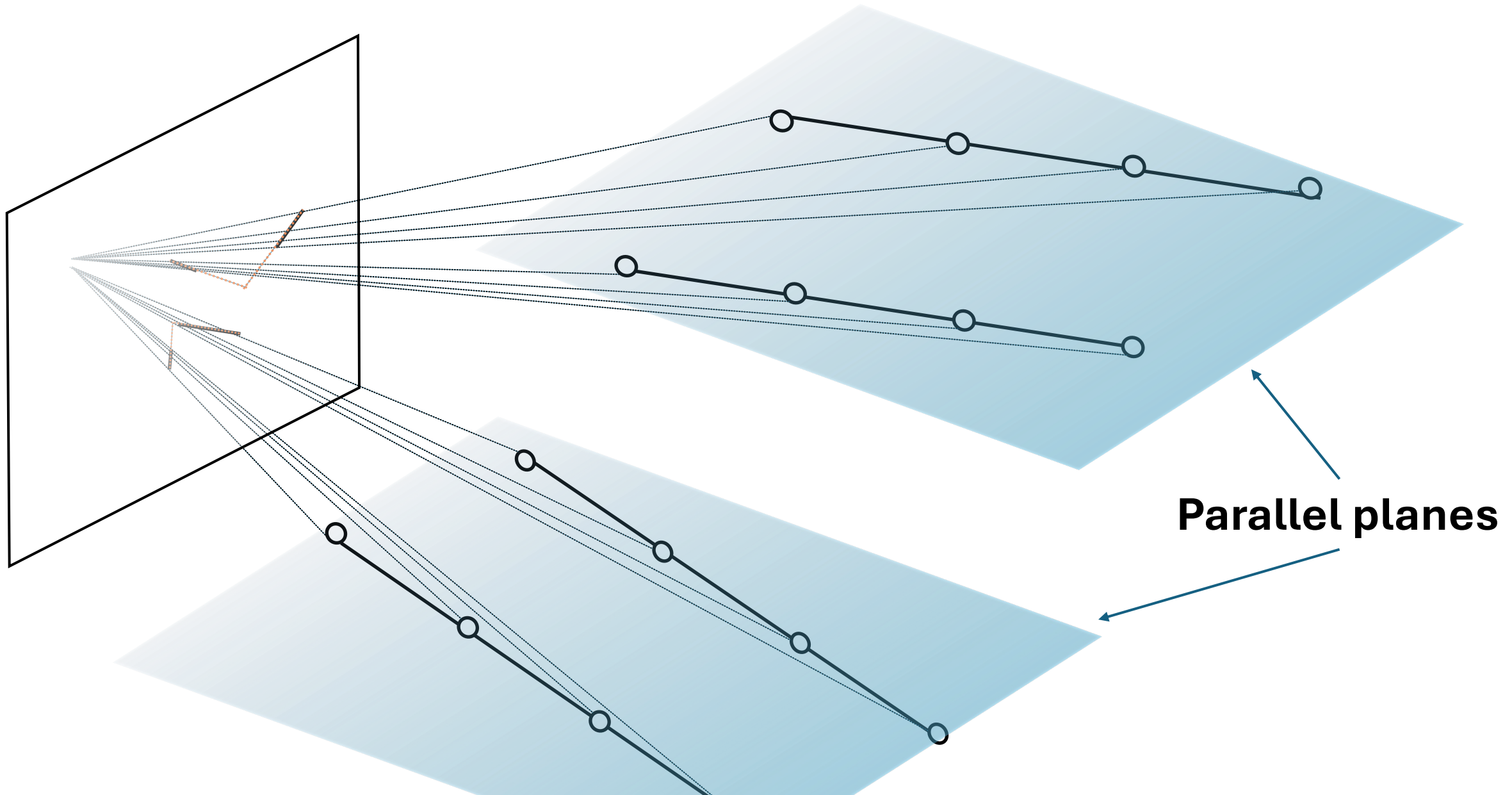


- The line through p_1 and p_2 is $l = p_1 \times p_2$

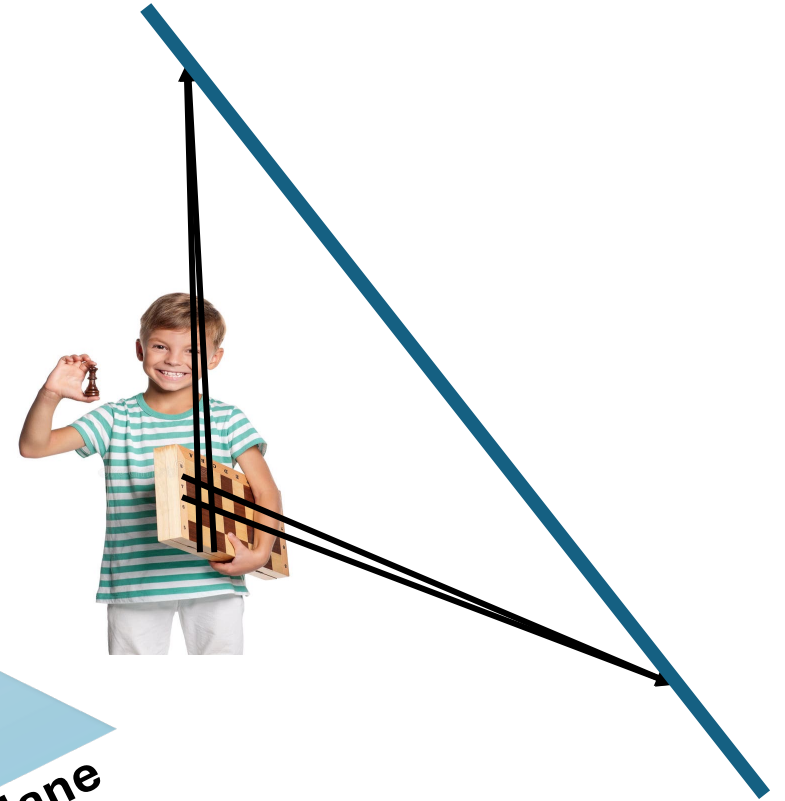
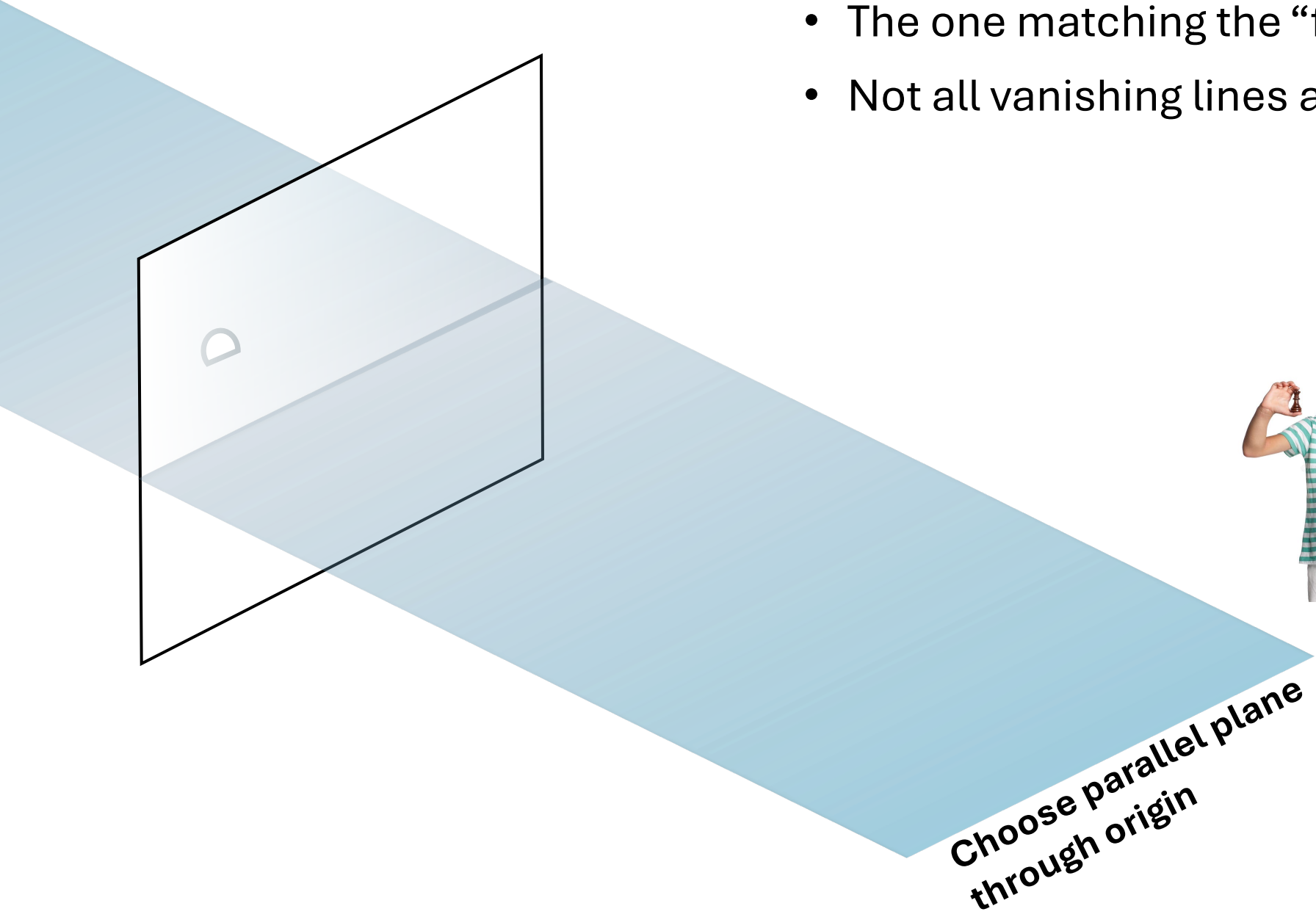
- The intersection of two lines $l_1^T p = 0$ and $l_2^T p = 0$ is $p = l_1 \times l_2$



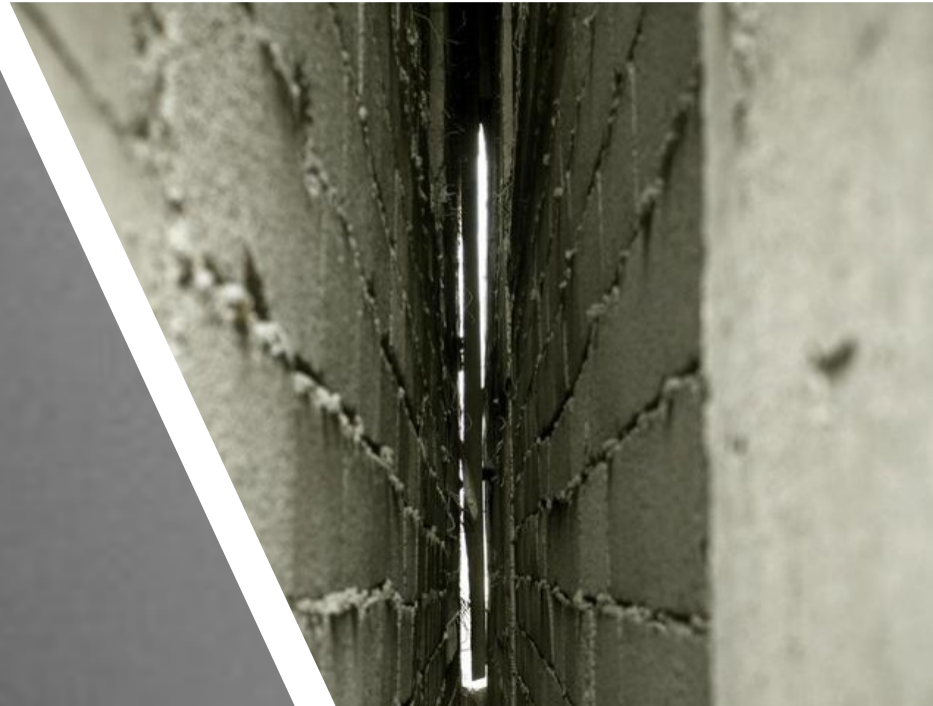
Vanishing line



- Each set of parallel planes has its vanishing lines
- The one matching the “floor” is called **Horizon**
- Not all vanishing lines are horizontal!

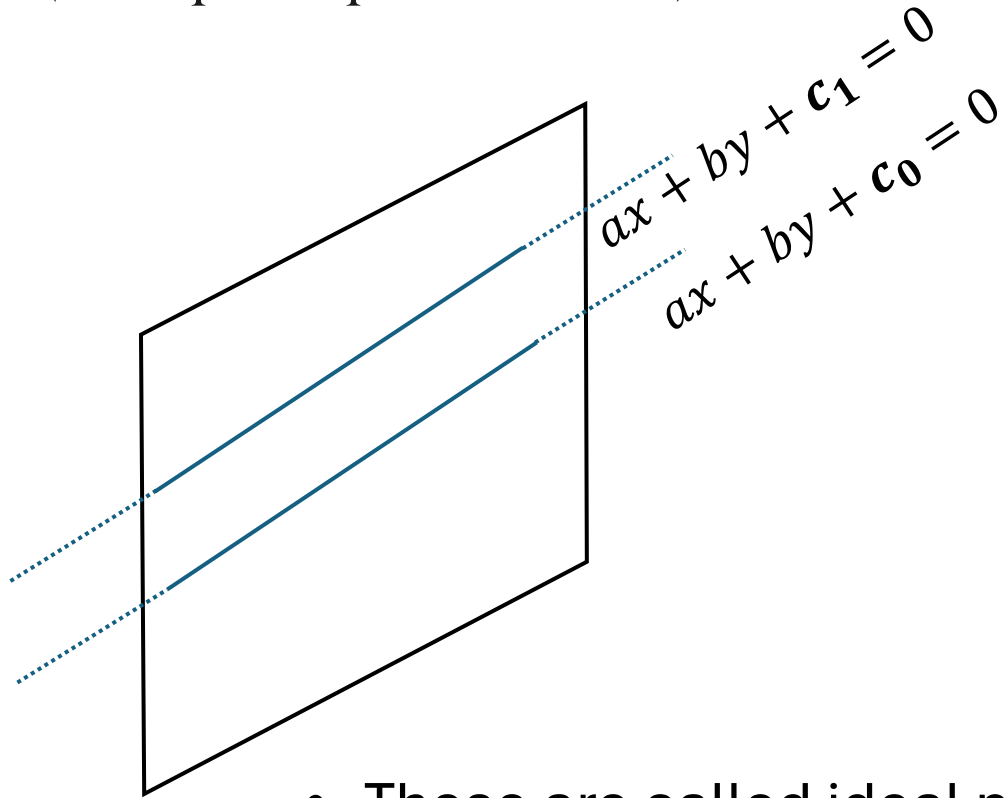


A set parallel lines intersects
at a single point.



What about parallel lines on the image plane?

(Or a plane parallel to it?)



$$l_0 \times l_1 = \begin{pmatrix} a \\ b \\ c_0 \end{pmatrix} \times \begin{pmatrix} a \\ b \\ c_1 \end{pmatrix}$$

$$= \underbrace{(c_1 - c_0)}_{const} \begin{pmatrix} b \\ -a \\ 0 \end{pmatrix}$$

- These are called ideal points.
- Not Euclidean.
- They make a line called “The line at infinity”.
- The line at infinity is indeed a line = a plane of rays through the origin (the $Z = 0$ plane)

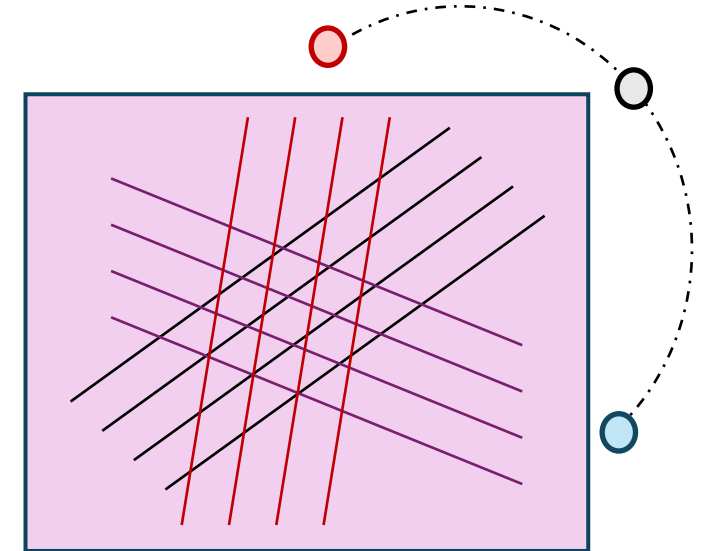
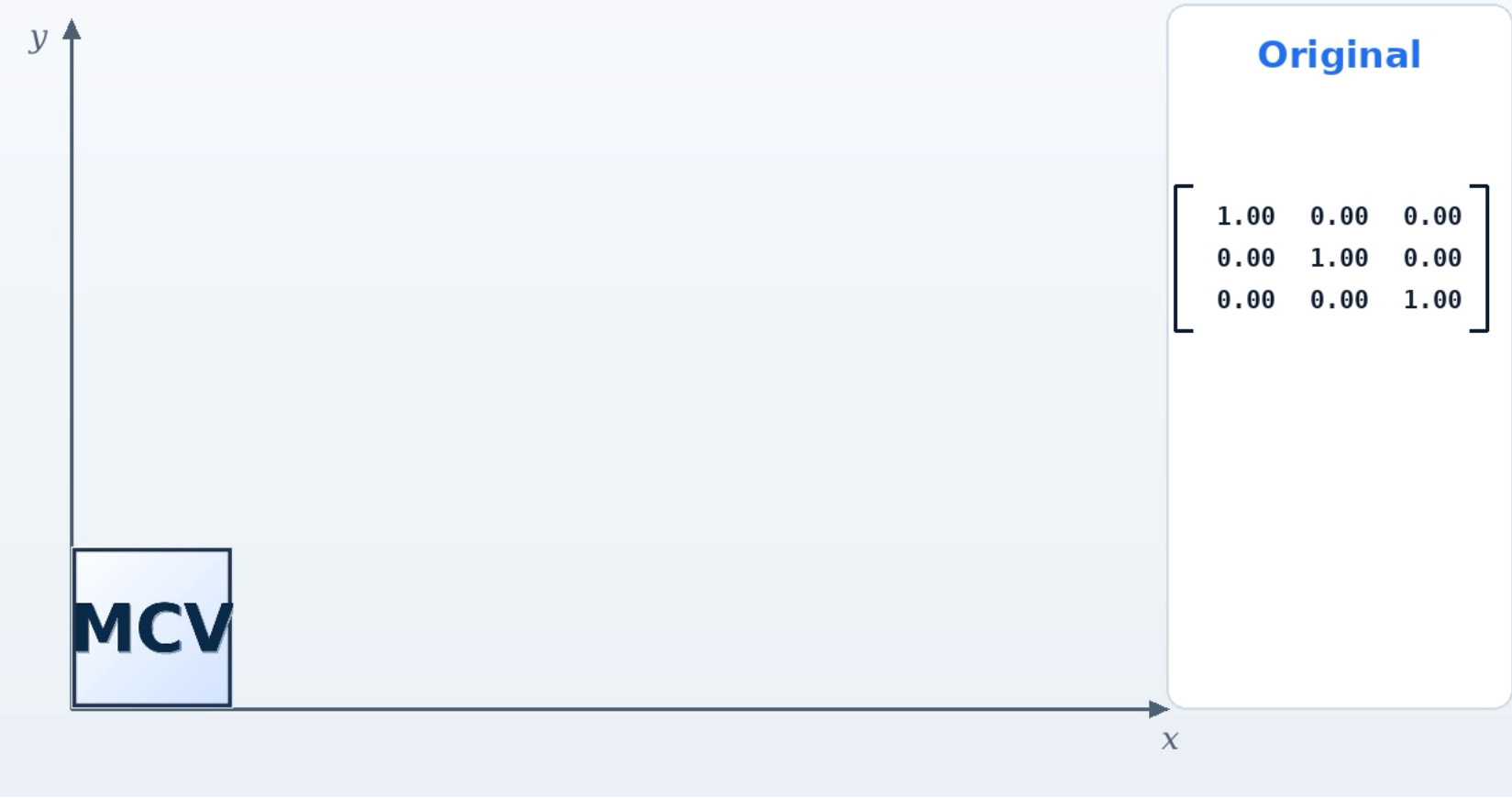


Figure credit: Ronen Basri

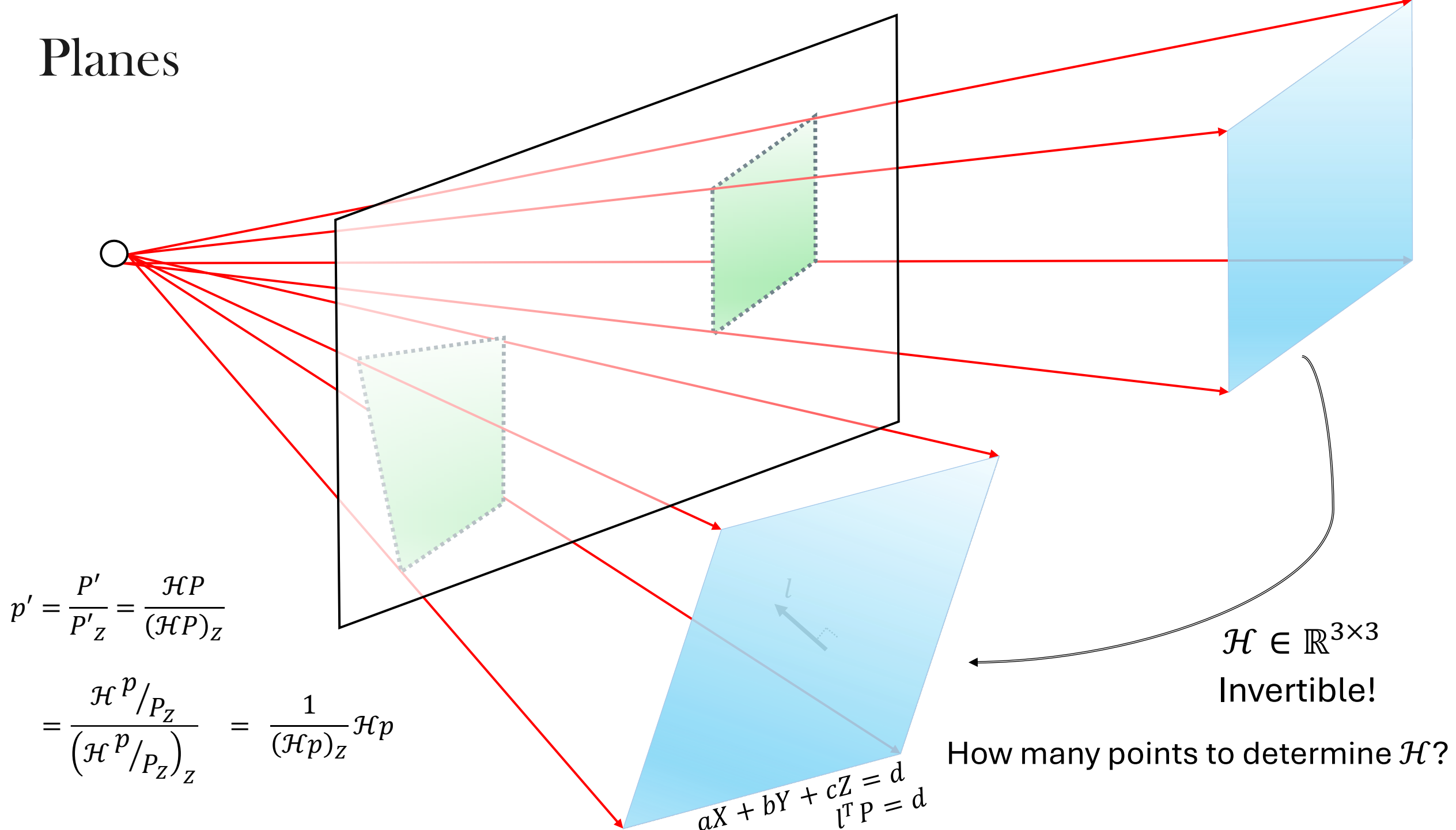
Transformations

Rigid	$\begin{pmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, lengths, area, parallelism
Similarity	$\begin{pmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, parallelism
Affine	$\begin{pmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves parallelism
Homography	$\begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix}$	Preserves cross ratio

The 2D Transformation Hierarchy



Planes



Extract homography

$$q = \frac{1}{(\mathcal{H}p)_z} \mathcal{H}p.$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \frac{1}{h_{31}x + h_{32}y + 1} \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + 1},$$

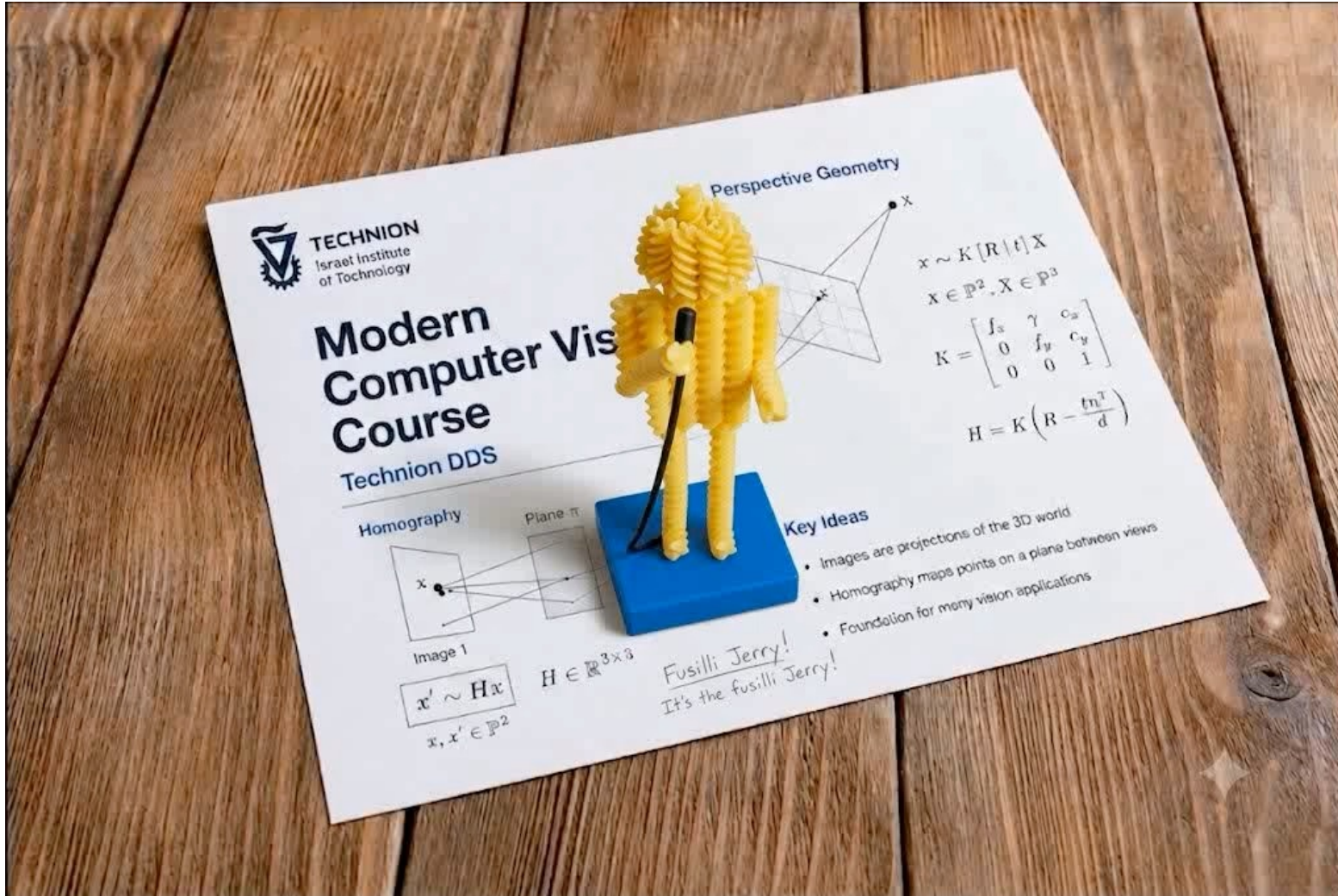
$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + 1}$$

$$h_{11}x + h_{12}y + h_{13} + h_{31}xx' + h_{32}yx' = -x'$$

×4 (x, y)
pairs

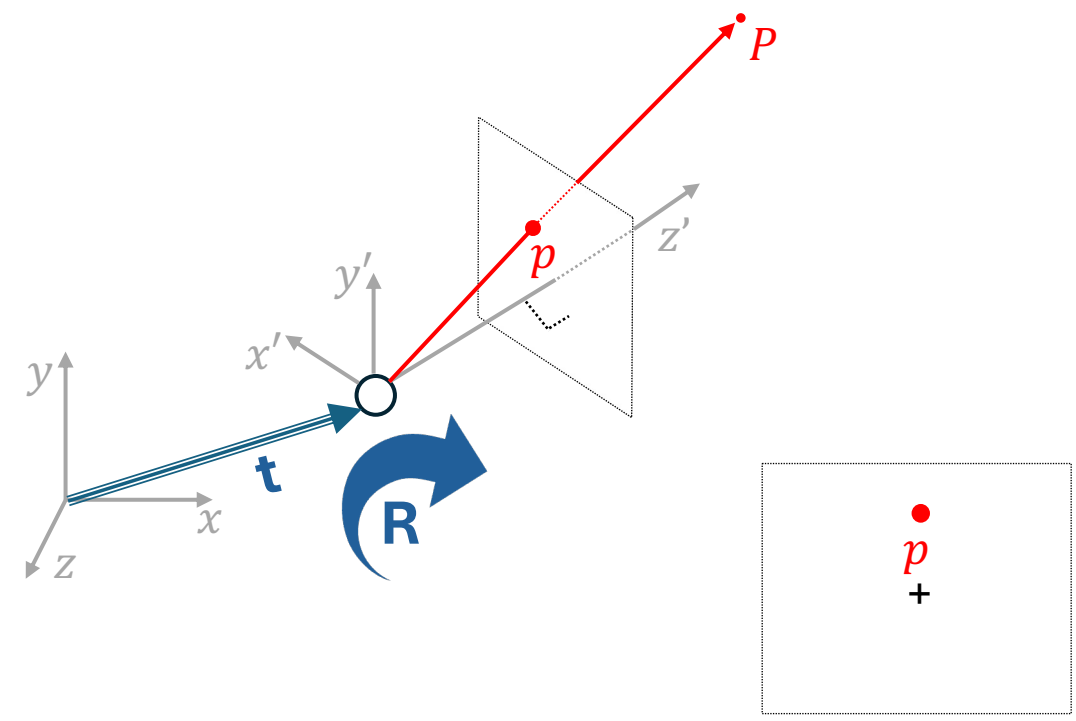


$$\begin{matrix} & \mathcal{M} & & \mathcal{H}_{flat} \\ \begin{pmatrix} x & y & 1 & 0 & 0 & 0 & xx' & yx' \\ 0 & 0 & 0 & x & y & 1 & xy' & yy' \\ & & & \bullet & & & & \\ & & & \bullet & & & & \\ & & & \bullet & & & & \end{pmatrix} & \begin{pmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{pmatrix} & = & \begin{pmatrix} -x' \\ -y' \\ \bullet \\ \bullet \\ \bullet \end{pmatrix} \end{matrix}$$



1. Find page corners A, B, C, D .
2. Set targets A', B', C', D' as:
 $(0,0), (0,1), (1,0), (1,1)$.
3. $\mathcal{H}_{flat} = \mathcal{M}^{-1} \begin{pmatrix} -x_A' \\ -y_A' \\ \vdots \end{pmatrix}$.
4. Warping: for each pixel in the **target**:
image $I' \begin{pmatrix} x \\ y \end{pmatrix} := I \left(\mathcal{H}^{-1} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right)$.
5. Use interpolation (grid-sample)

Camera matrix



$$R = R_z(\psi)R_x(\phi)R_y(\theta)$$

$$= \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \psi \cos \theta - \sin \phi \sin \psi \sin \theta & -\sin \psi \cos \phi & \sin \theta \cos \psi + \sin \phi \sin \psi \cos \theta \\ \sin \psi \cos \theta + \sin \phi \sin \theta \cos \psi & \cos \phi \cos \psi & \sin \theta \sin \psi - \sin \phi \cos \psi \cos \theta \\ -\sin \theta \cos \phi & \sin \phi & \cos \phi \cos \theta \end{bmatrix}$$

$$P' = RP + t$$

$$\begin{pmatrix} X' \\ Y' \\ Z' \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$P' = [R|t]P$$

$$p' = \frac{1}{Z[R|t]P} K[R|t]P$$

$$\begin{pmatrix} x \\ y \\ f \end{pmatrix} \propto \underbrace{\begin{pmatrix} fm_x & \gamma & x_0 \\ 0 & 0 & y_0 \\ 0 & 0 & 1 \end{pmatrix}}_K \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Skew

Sensor center

Focal dist.

Pix density

$$p' \propto K[R|t]P$$

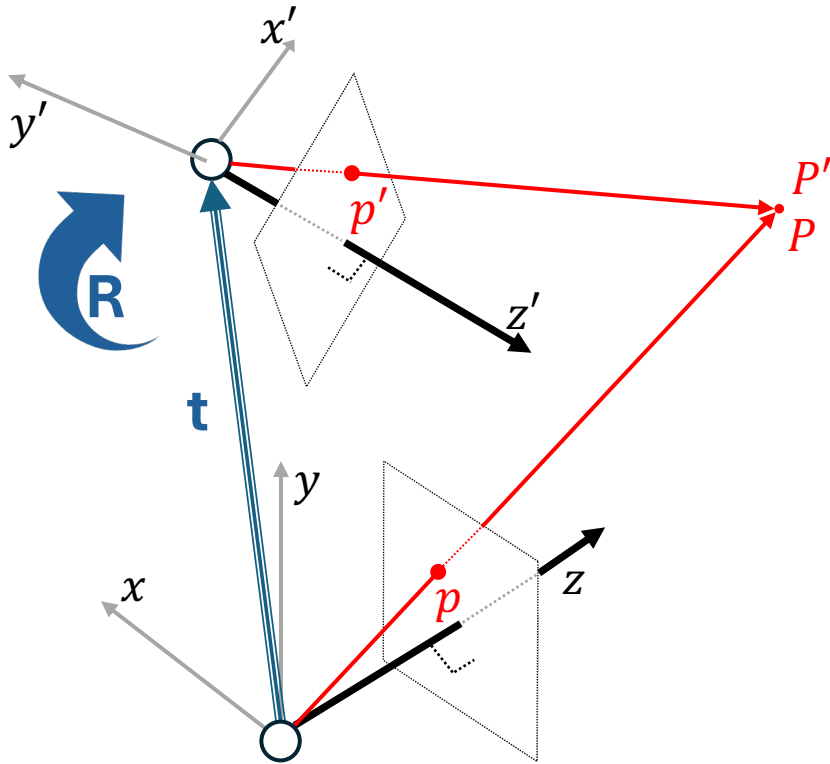
3D reconstruction from stereo vision setup

$$p' = \frac{1}{Z_{[R|t]_P}} K[R|t]P$$

Goal: Depth $\rightarrow$ 3D reconstruction.

$$P' = RP + t$$

For known K and normalized p, p' : $\mathbf{Z}' p' = \mathbf{Z} R p + t$



Essential matrix

If we recover K, R, t we can get depth of P :

Calibration

$$\mathbf{Z}' \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \mathbf{Z} R \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix}$$

3 equations, 2 unknowns.

For unnormalized coords:

$$\mathbf{Z}' K'^{-1} p' = \mathbf{Z} K^{-1} R p + t$$

A note about calibration

We don't get in to details, but here is the gist:

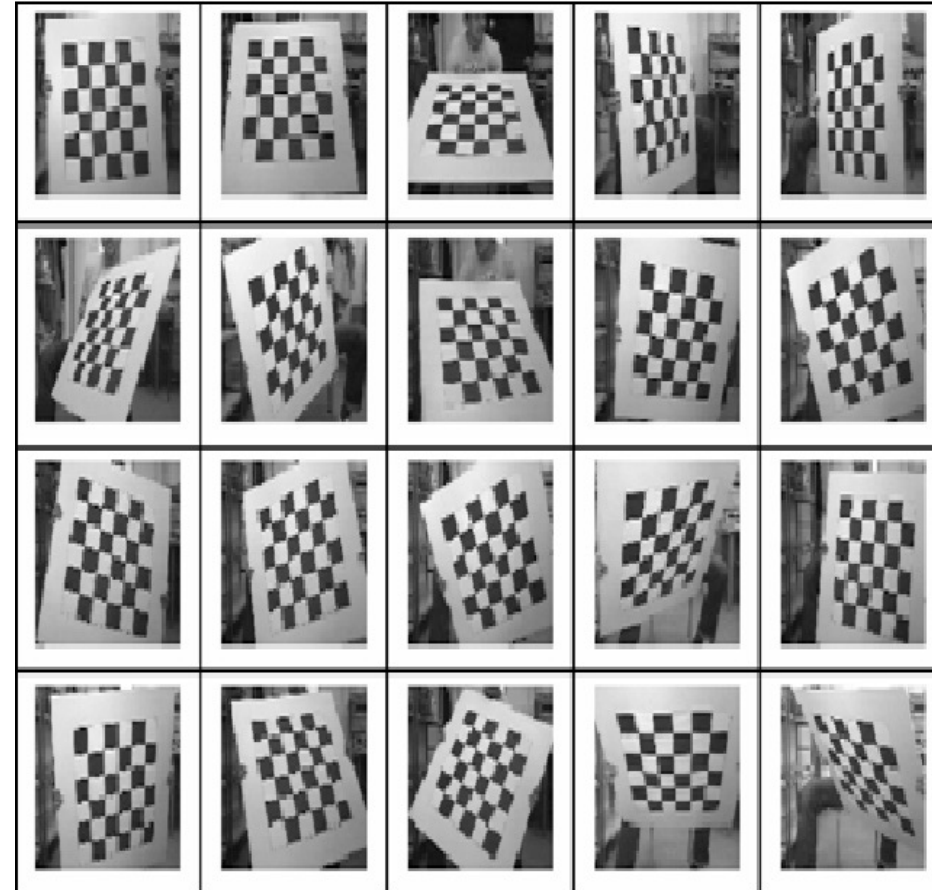
1. Have a flat item with known sizes.
2. Take many pics.
3. Get "CamScanner" homographies for all. So move to $Z=0$.
4. Use cam mat with $Z=0$ and known x, y, X, Y :

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = K \begin{bmatrix} r_1 & r_2 & t \end{bmatrix} \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$

5. We don't need $\begin{bmatrix} r_1 & r_2 & t \end{bmatrix}$, only K . We use the properties of R as a trick: $r_1 \perp r_2$, $\|r_1\| = \|r_2\|$ to get two equations per image:

$$r_1^T K^{-T} K^{-1} r_2 = 0, r_1^T K^{-T} K^{-1} r_1 = r_2^T K^{-T} K^{-1} r_2$$

6. When done we have $K^{-T} K^{-1}$. To factor out K we need to use Cholesky decomposition.



Multiple views of planar scenes

Camera can change position and orientation (and intrinsics).

$$P' = RP + t$$

$$P' = RP + t \cdot 1$$

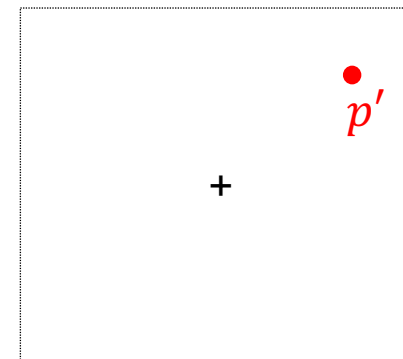
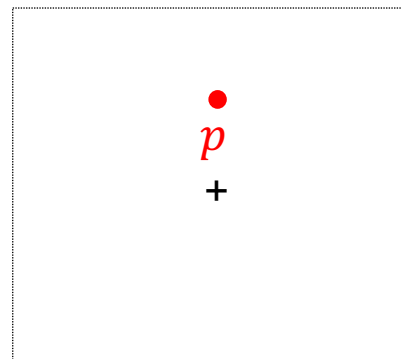
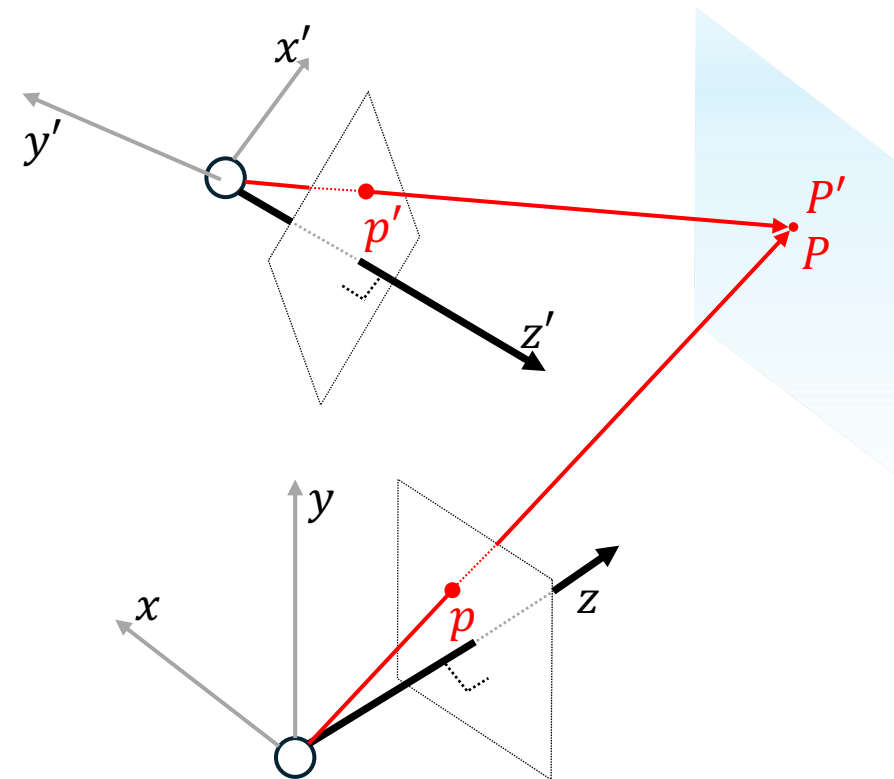
$$P' = RP + t \cdot \frac{l^T P}{d}$$

$$P' = \left(R + t \cdot l^T / d \right) P$$

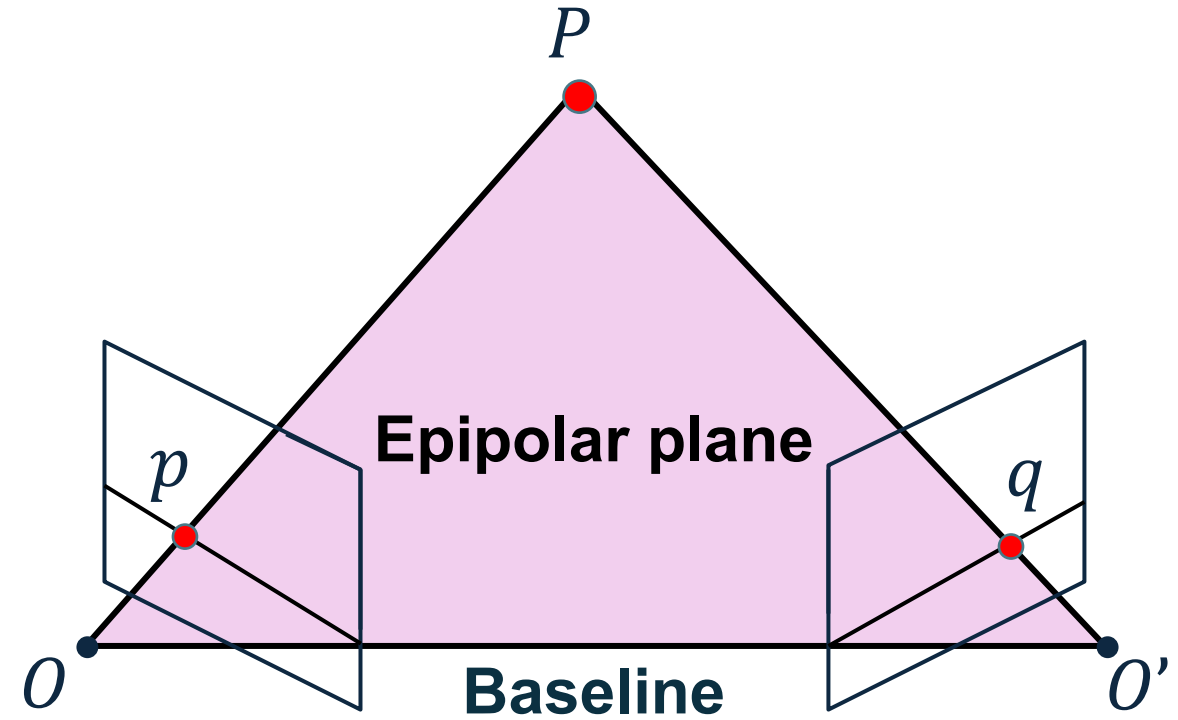
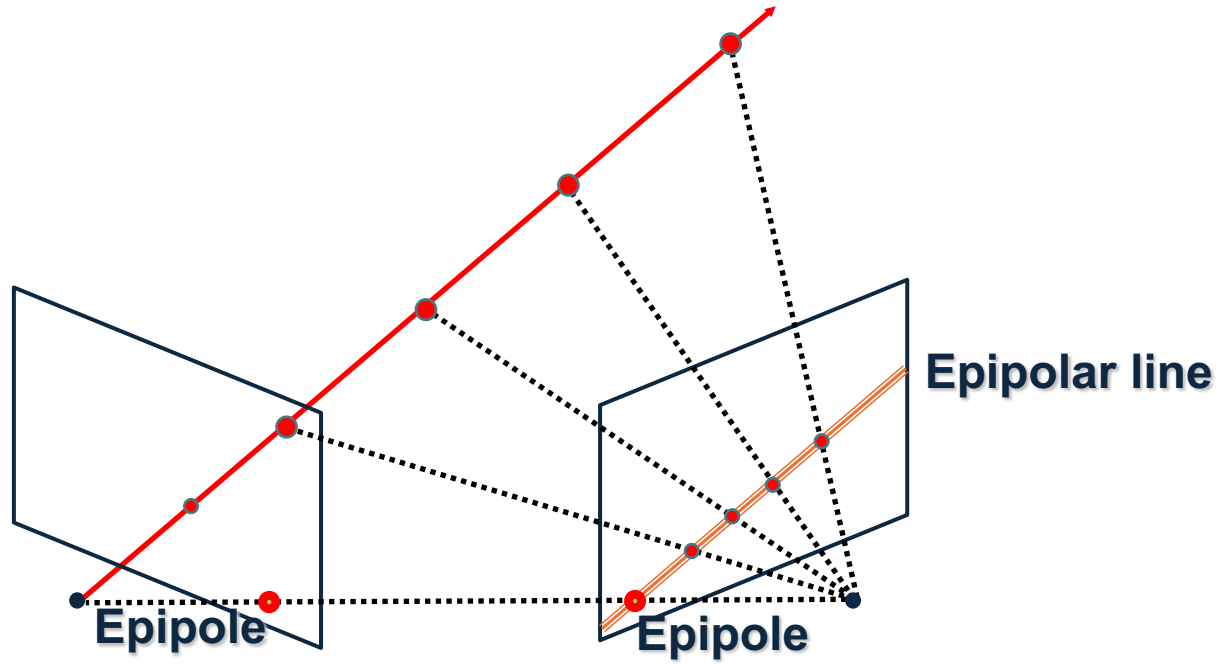
$$p' \propto \underbrace{\left(R + t \cdot l^T / d \right)}_{\mathcal{H}} p$$

$$P \text{ is on the plane:}$$

$$l^T P = d \Rightarrow \frac{l^T P}{d} = 1$$

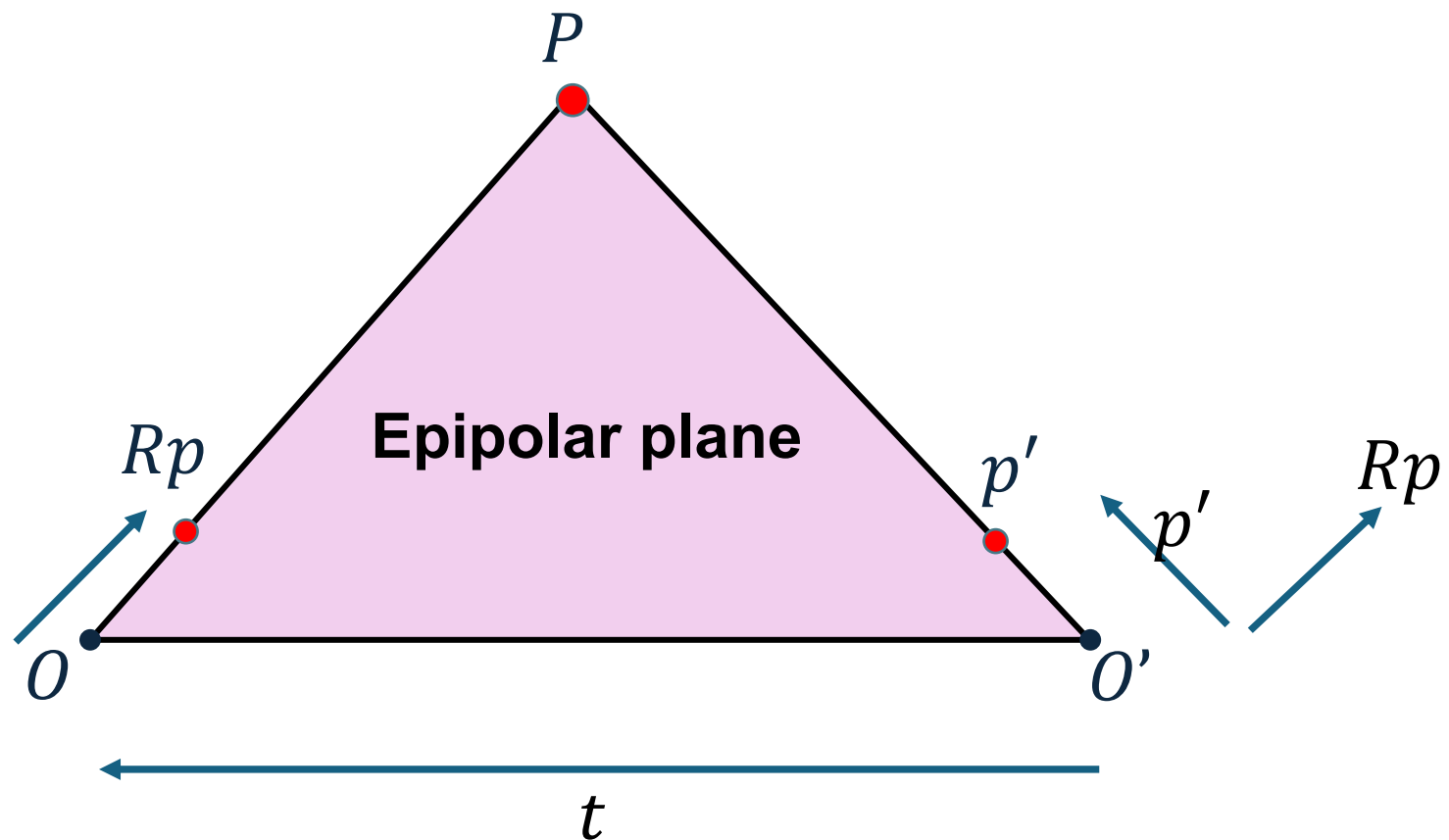


Epipolar lines



Essential matrix

(Assuming $K = I$)



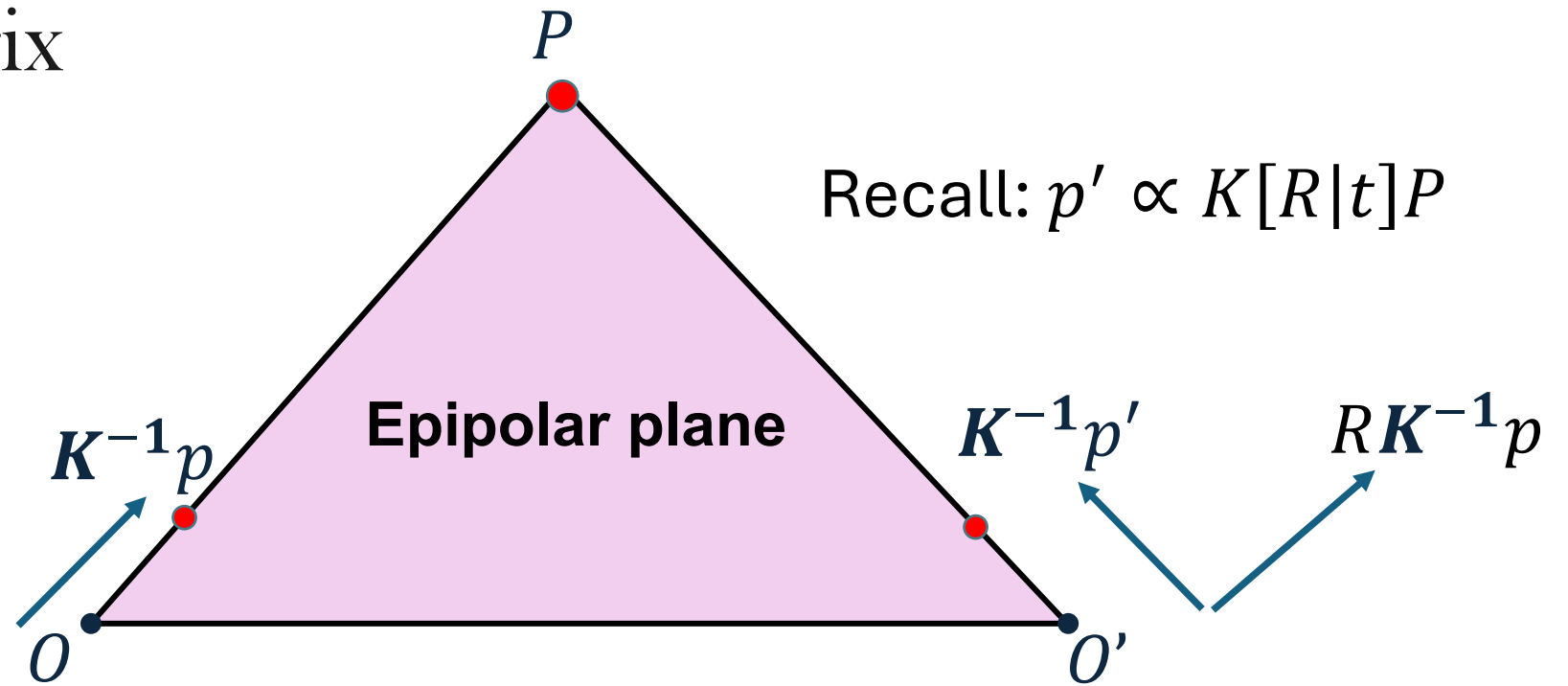
Coplanar in O' coords: $p'^T(t \times Rp') = 0$

$$\underbrace{p'^T[t]_{\times}}_{\mathbf{E}} Rp = 0$$

$$p'^T \mathbf{E} p = 0$$

Fundamental matrix

Camera is uncalibrated



Recall: $p' \propto K[R|t]P$

$p^T E p' = 0$ only holds for $K = I$, calibrated camera.

Coplanar in O' coords: $p^T K^{-T} (t \times R K^{-1} p') = 0$

$$p'^T \underbrace{K^{-T} [t]_{\times} R K^{-1}}_{\mathbf{F}} p = 0$$

$$\mathbf{F} = K^{-T} \mathbf{E} K^{-1}$$

$$p'^T \mathbf{F} p = 0$$

Extract F / E

For each pair of corresponding points: $p'^T F p = 0$

$$(x \ y \ 1) \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{pmatrix} \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} x'x & x'y & x' & y'x & y'y & y' & x & y & 1 \\ & & \bullet & & & & & & \\ & & \bullet & & & & & & \\ & & \bullet & & & & & & \end{pmatrix} \begin{pmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{pmatrix} = 0$$

Q: How do you avoid trivial solution? A: Singular vec matching minimal singular val!

Note: After reshaping back to f , we need to enforce rank 2. SVD-> set singular vals: (1, 1, 0)

Extract R , t

$$E = [t]_{\times} R = U \Sigma V^T$$

$$E^T t = -R^T(t \times t) = 0 \rightarrow t = \text{left null vector} \rightarrow \mathbf{t} \propto \mathbf{u}_3 \text{ (the epipole)}$$

3rd sing val is 0

2 important facts about R :

1. R is a rotation so all singular vals are 1. But E has vals $(1, 1, 0)$.
2. R returns vecs orthogonal to E : $(Rx)^T(Ex) = 0$ because $Ex = t \times Ex$.

$$R = U \begin{bmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ -\mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} V^T, \quad \mathbf{t} \propto \mathbf{u}_3$$

3D reconstruction from stereo vision pipeline:

1. **[CORRESPONDENCES]**: A whole story by itself...

2. **[EPIPOLAR GEOMETRY ESTIMATION]**: ≥ 8 points to get F by $p'^T F p = 0$.

RANSAC

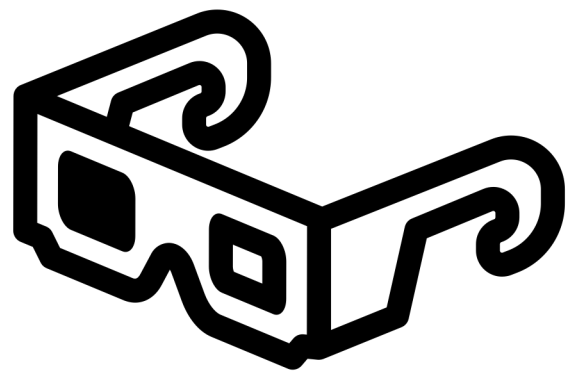
3. **[CALIBRATION]**: to get K, E : $E = K^T F K$.

4. **[CAMERA POSE]**: Extract R, t of E : $R = U \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} V^T, \quad t \propto u_3$.

5. **[TRIANGULATION]**: Per point 3 eqs 2 unknowns: $Z' K'^{-1} p' = Z K^{-1} R p + t$

* Choose positive depth result

Demo:



Thank you!

Next week:

Motion

